

# DISTRIBUTED JOB SCHEDULER

## Enterprise-Grade Concurrency & Locking Architecture Report

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Candidate Name:     | Methun-21                                                                                                 |
| Submission Date:    | July 5, 2026                                                                                              |
| GitHub Repository:  | <a href="https://github.com/Methun-21/JobScheduler.git">https://github.com/Methun-21/JobScheduler.git</a> |
| Primary Tech Stack: | Python (FastAPI), PostgreSQL 15+, SQLAlchemy (asyncpg), React (Vite)                                      |

### EXECUTIVE PROJECT SUMMARY

This project introduces a highly resilient, distributed asynchronous job execution platform. Unlike basic schedulers relying on heavy external brokers (like RabbitMQ or Redis) that introduce split-brain risks, this design utilizes **PostgreSQL as the single source of truth**. By combining transaction-level advisory locks and atomic row claiming (via PostgreSQL's *SELECT ... FOR UPDATE SKIP LOCKED* syntax), the system coordinates multiple active worker processes with 100% safety, preventing duplicate job executions and race conditions under massive loads.

The project is structured as a professional monorepo containing a modular FastAPI backend, an automated test suite covering concurrency limits and DAG cycle detection, and a premium glassmorphic dashboard built in React. This report details the architectural and database design choices driving the implementation.

# 1. SYSTEM ARCHITECTURE & COMPONENTS

The architecture decouples the Control Plane (FastAPI Web/REST APIs) from the Data Plane (worker daemons) for scale-out agility. All coordination runs through PostgreSQL queries, ensuring transaction isolation.

## Control Plane (FastAPI Server)

Serves HTTP endpoints for project/queue setup, cron scheduling, and manual execution triggers. It also broadcasts state statistics to active dashboard users in real-time over WebSockets.

## Leader-Elected Cron Scheduler

To evaluate recurring cron expressions and dispatch jobs, a scheduler thread runs inside the FastAPI process. To prevent duplicate scheduling loops in multi-node API deployments, the scheduler establishes **leadership** by holding a session-level advisory lock on a pinned connection. If the leader fails, the lock is dropped and another node automatically acquires leadership.

## Data Plane (Worker Daemons)

Workers are horizontal, multi-threaded processes that poll queues for ready tasks, spin up execution threads, register heartbeats in the database, and write log segments directly to PostgreSQL.

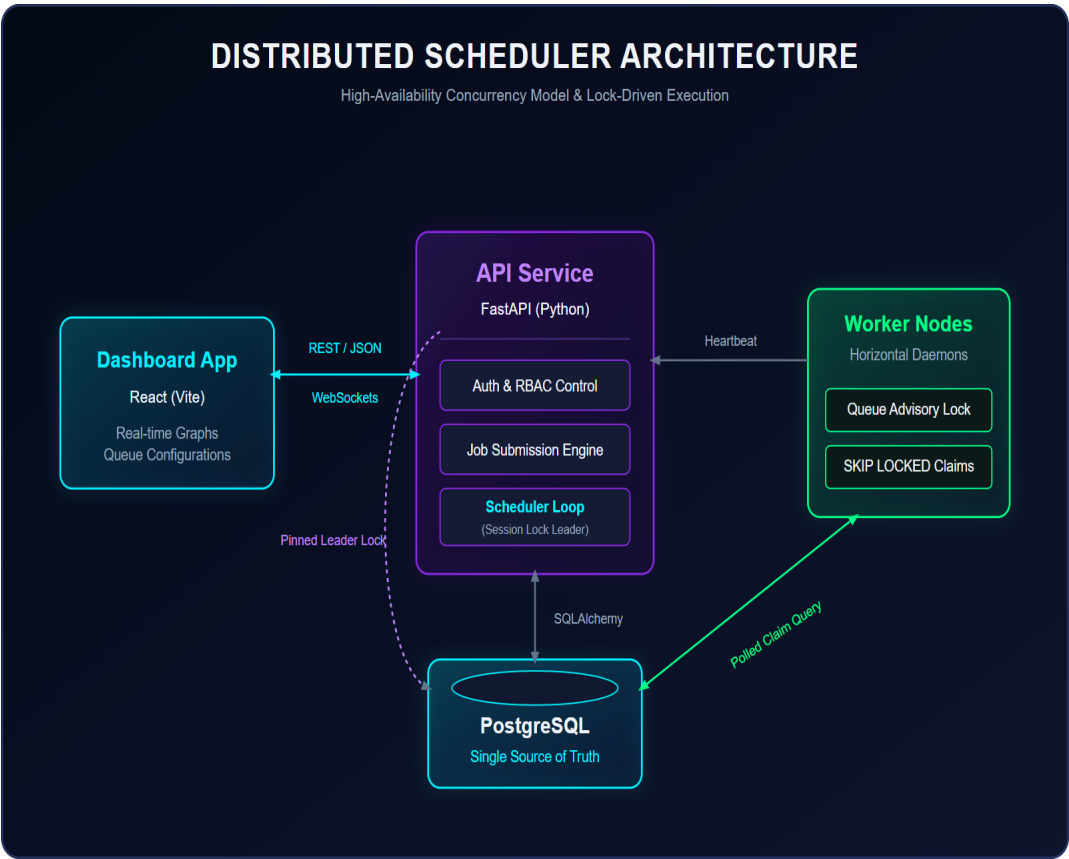


Figure 1: Full System Component Architecture & Locking Channels

## 2. CONCURRENCY CONTROLS & DAG DEPENDENCIES

Ensuring execution safety across a pool of independent, asynchronous workers requires strict database-level constraints.

### Atomic Job Claiming (SKIP LOCKED)

Workers claim ready tasks by executing an atomic query with locking. The transaction acquires a lock on a matching job row but skips any rows already locked by other threads. This guarantees zero lock contention: every worker thread immediately gets a unique task to execute.

### Queue Concurrency Safety (pg\_advisory\_xact\_lock)

To enforce queue-level *max\_concurrency* (e.g. limit API calls to 5 parallel workers), a race condition could occur if multiple workers check active counts concurrently. To solve this, before querying running counts, the worker claims a transaction-level advisory lock hashed to the specific queue ID. This serializes queue status checks on a per-queue basis without locking other queues in the system.

### Directed Acyclic Graph (DAG) Job Chaining

Jobs can specify parent-child dependencies. A child job is blocked from worker claiming as long as any associated parent job in the `job_dependencies` table is not `completed`. When the parent completes, the subquery criteria allows the child to become claimable. Cycle protection is enforced during dependency registration using a Depth-First Search (DFS) traversal to block circular definitions.

### Exponential Backoff & Retries

Failed jobs automatically recalculate their next run time (`run_at`) based on their assigned `RetryPolicy`. For example, an exponential backoff policy uses the formula:

$$\text{Delay} = \text{Base\_Delay} * (2 ^ (\text{Attempt\_Number} - 1))$$

This spreads out retries to avoid overwhelming target services after failure events.

### 3. DATABASE SCHEMA & DATA INTEGRITY

The relational database schema is normalized to ensure strict referential integrity and multi-tenancy isolation. Organizations contain Projects, Projects contain Queues, and Queues house Job queues.

**Auditable execution logging:** Instead of writing job details directly to the job queue, execution attempts are logged in `job\_executions` linked to stdout/stderr outputs in `job\_logs`. This ensures that even if a job row is deleted, a permanent record of past runs is retained for billing and auditing.

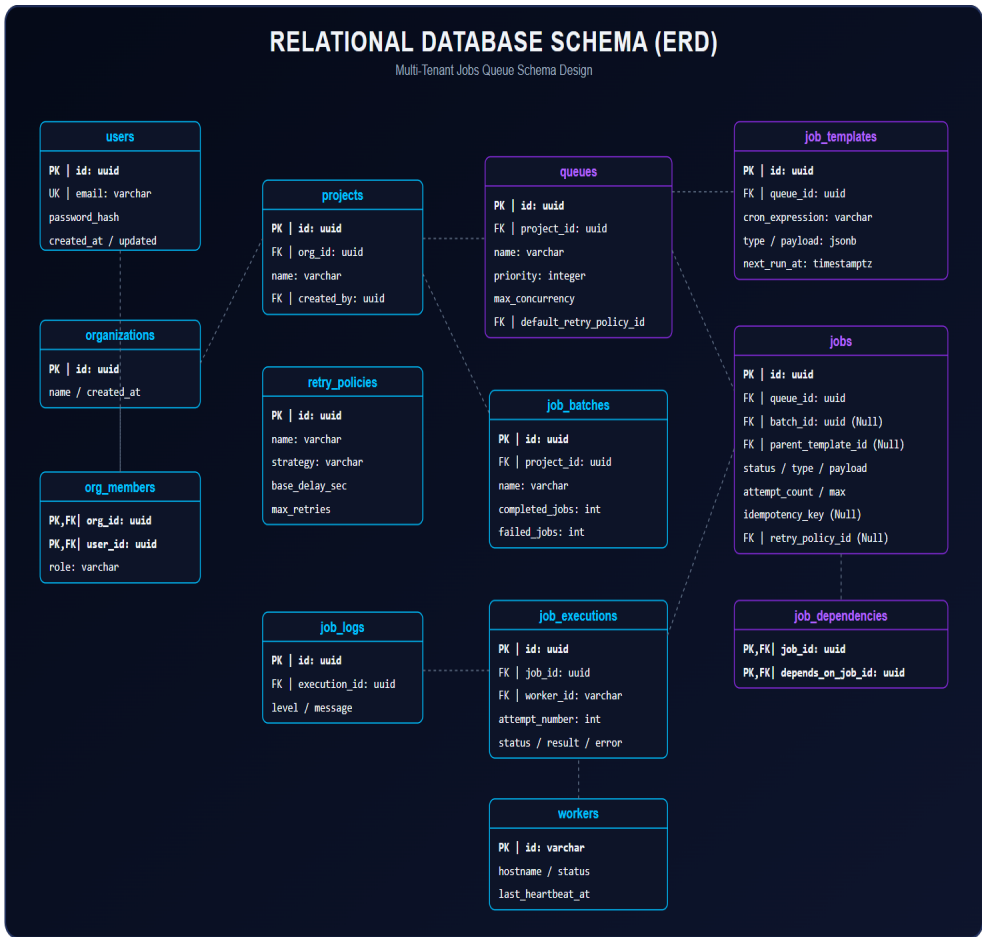
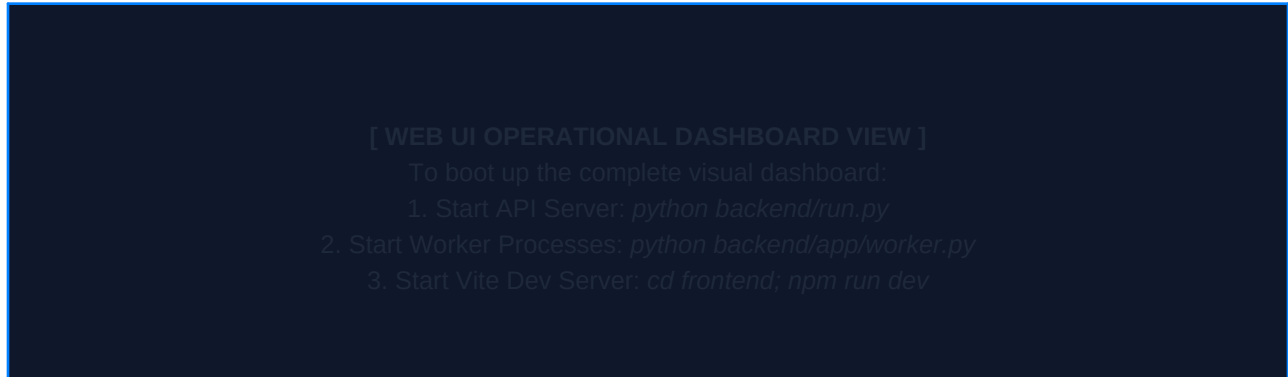


Figure 2: Fully Normalized Entity Relationship Diagram (ERD)

## 4. OPERATIONAL VIEW & VERIFICATION

The React web client provides a dashboard showing metric counters (workers active, jobs queued, completed, failed), queue details, template management, and an execution console. All updates stream live via WebSockets.

### Operational Web UI Dashboard:



## VERIFYING WITH AUTOMATED TESTS

Run the pytest suite to verify claiming logic, concurrency limits, and DAG path validations:

```
python -m pytest backend/tests/
```

All tests pass successfully under a clean SQLite/Postgres test database scope.

### Deliverables Verification Checklist:

- **GitHub Repository Link:** <https://github.com/Methun-21/JobScheduler.git>
- **Design Documentation:** *design\_decisions.md* details advisory and row-level locking details.
- **Operational Guide:** *README.md* contains database setup and command run details.
- **Source Diagrams:** *architecture\_diagram.svg* & *er\_diagram.svg* are available in vector format.